Advancing machine learning for protein structure prediction in drug discovery

Specific Aims

Aim 1: Advance machine learning for protein structure prediction in drug discovery via a characterization-focused strategy.

Aim 2: Advance machine learning for protein structure prediction in drug discovery via a characterization-focused strategy.

Background

Our preliminary data suggest a tractable path toward measurable improvement. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Findings will be disseminated through peer-reviewed publication and open data release. Prior work has established a partial understanding, yet key gaps remain.

Approach

This proposal addresses machine learning for protein structure prediction in drug discovery, a problem with substantial significance. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Findings will be disseminated through peer-reviewed publication and open data release. We will assemble a multidisciplinary team with complementary expertise.

Innovation

Our preliminary data suggest a tractable path toward measurable improvement.

Investigator Context

The PI's first independent award; strong mentorship is in place. The R2 host institution offers focused departmental resources and regional research partnerships, backed by a growing institutional research base.

Budget Justification

Total requested support is $575,604 over 3 years at a R2 institution, covering personnel, materials, and dissemination.

References

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7. [7] https://doi.org/10.9525/2hn5f62d
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